

CMSC216: Bonus Review 1A

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Bonus Review Rules

- ▶ 2 Questions will be shown with about 5min per question, 10min total, time limit enforced on Gradescope Quiz
- ▶ Each correct answer worth 1 Engagement Point
- ▶ Cooperation is allowed and encouraged within your discussion section - *talk to your neighbors, form a class consensus!*
- ▶ Scores posted after all sections have completed bonus review, likely the following day
- ▶ Review is Open Resource just like the exam:
<https://www.cs.umd.edu/~profk/216/exam-rules.pdf>

Staging

- ▶ Open up the Gradescope Review Quiz for the day
- ▶ Once started, the quiz closes after 10min

Get everybody and their stuff together

Okay...



Question 1

In Project 1, which of the following functions would correctly sum the total valid Votes tracked by a Tally?

Mark ALL in (A)-(D) that are correct

///// (A) /////

```
int tally_total_votes_A(tally_t *t){
    int total = 0;
    for(int i=0; i < t.candidate_count; i++){
        total += t.candidate_vote_counts[i];
    }
    return total;
}
```

///// (C) /////

```
int tally_total_votes_C(tally_t *t){
    int total = 0;
    for(int i=0; i < t->candidate_count; i++){
        total += t->candidate_vote_counts[i];
    }
    return total;
}
```

///// (B) /////

```
int tally_total_votes_B(tally_t *t){
    int total = 0;
    for(int i=0; i < t->candidate_count; i++){
        total += sizeof(t->candidate_votes[i]);
    }
    return total;
}
```

///// (D) /////

```
int tally_total_votes_D(tally_t *t){
    int total = 0;
    for(int i=0; i < t->candidate_count; i++){
        vote_t *vote = t->candidate_votes[i];
        while(vote != NULL){
            total++;
            vote = vote->next;
        }
    }
    return total;
}
```

Question 2

Study to code below and the associated Valgrind output.

```
1  typedef struct {
2      float *data; int count; float total;
3  } samp_t;
4
5  void samp_normalize(samp_t *samp){
6      for(int i=0; i<samp->count; i++){
7          samp->data[i] /= samp->total;
8      }
9  }
10
11 int main(){
12     ...
13 }
```

```
>> gcc -g sample.c
>> valgrind ./a.out
==973== Invalid read of size 4
==973==    at 0x4116F: samp_normalize (sample.c:6)
==973==    by 0x41196: main (sample.c:12)
==973== Address 0x8 is not stackd, mallocd or
==973==    (recently) freed
==973==
==973== Process terminating with default action of
==973==    signal 11 (SIGSEGV):
```

Which of the following is likely contributing to the bug?

- ▶ (A) Likely the data field is NULL
- ▶ (B) Likely the data field has too little memory allocated
- ▶ (C) Likely the count field is uninitialized
- ▶ (D) Likely the total field has too little memory allocated
- ▶ (E) Likely the samp variable is NULL
- ▶ (F) Likely the i variable is NULL